Household Carbon Footprint & Target Tracker

Architectural Style Justification

We chose Layered Architecture for the Household Carbon Footprint & Target Tracker system.

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (User Interface Component serving both the Resident and Sustainability Coach), a Business Layer (Consumption & Target Manager, CO2e Conversion Service, and Notification & Coach Review components), and a Data Layer (Database Component). Each layer only communicates with the layer directly below it through well-defined interfaces, keeping the system's core emissions-calculation and target-tracking logic cleanly separated from both the UI and the data store.

Two Reasons for This Choice

- Moderate, well-understood scope: this is a single-household analytics tool with a small, stable set of functions (log consumption, compute CO2e, set targets, track progress, notify, and support coach review) rather than dozens of independently scaled services. A Layered architecture matches this scope without the operational overhead of Microservices (separate deployments, service discovery, network calls between every component) or the scalability and fault-tolerance weaknesses of a plain Client-Server design.
- Swappable calculation logic: EPA conversion factors are likely to be revised periodically, and the CO2e Conversion Service sits in its own component behind a clearly defined Conversion Interface. Because the Business Layer only depends on that interface and not on how the conversion is implemented internally, the conversion logic can be updated or replaced without touching the Consumption & Target Manager, the UI, or the database schema.

Security Advantage

Because the Presentation Layer never talks to the Database Component directly, it cannot accidentally issue a raw query that crosses household boundaries. All data access is funnelled through the Consumption & Target Manager's Database Interface, giving the system a single, auditable choke point where household-scoped access control (NFR-002) can be enforced consistently, rather than having to re-implement that check in every UI screen.

Performance Benefit

Keeping the CO2e Conversion Service and the Notification & Coach Review component as separate, in-process business-layer components (rather than remote microservices) avoids network round-trip latency on the hot path of logging a month's consumption. This directly supports NFR-001's sub-200ms chart-rendering requirement, since the data the chart needs has already been computed and persisted synchronously within the same layer, rather than waiting on a cross-service call at render time.